MA677 Final Paper Figures

Benoit (2011): Linear Regression Models with Logarithmic Transformations

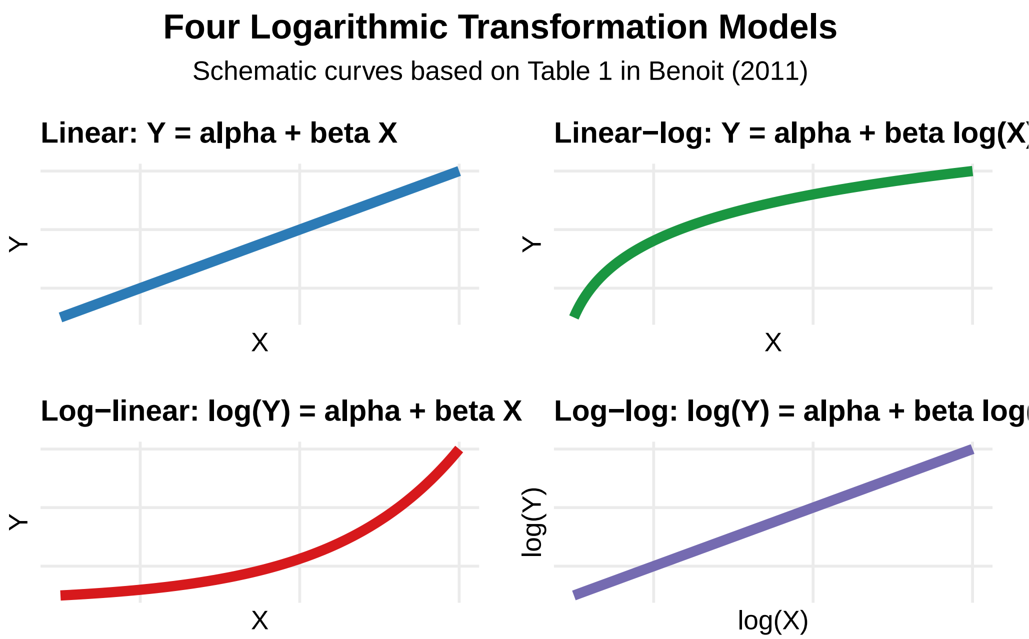
```r
library(ggplot2)
library(patchwork)
library(scales)
library(grid)

set.seed(7)

out_dir <- "figures"
dir.create(out_dir, showWarnings = FALSE)

coef_linlog_intercept <- -24.42095
coef_linlog_slope     <- 10.43004

coef_loglin_intercept <- 4.630287
coef_loglin_slope     <- 0.052709

coef_loglog_intercept <- 7.088676
coef_loglog_slope     <- -0.4984531

theme_paper <- theme_minimal(base_size = 11) +
  theme(
    plot.title = element_text(face = "bold", size = 11),
    plot.subtitle = element_text(size = 9.5),
    plot.caption = element_text(size = 8.5, hjust = 0),
    panel.grid.minor = element_blank(),
    axis.title = element_text(size = 10),
    plot.margin = margin(8, 8, 8, 8)
  )

blue   <- "#2c7bb6"
green  <- "#1a9641"
```

```r
red    <- "#d7191c"
purple <- "#756bb1"

norm01 <- function(v, lo = 1, hi = 6) {
  (v - min(v)) / (max(v) - min(v)) * (hi - lo) + lo
}
```

Figure 1

```r
x <- seq(1, 20, length.out = 200)

df_linear <- data.frame(x = norm01(x), y = norm01(x))
df_linlog <- data.frame(x = norm01(x), y = norm01(log(x)))
df_loglin <- data.frame(x = norm01(x), y = norm01(exp(x / 6)))
df_loglog <- data.frame(x = norm01(log(x)), y = norm01(log(x)))

make_shape_plot <- function(df, title, col, xl = "X", yl = "Y") {
  ggplot(df, aes(x, y)) +
    geom_line(color = col, linewidth = 1.8) +
    labs(title = title, x = xl, y = yl) +
    theme_paper +
    theme(axis.text = element_blank(), axis.ticks = element_blank())
}

p1a <- make_shape_plot(df_linear, "Linear: Y = alpha + beta X", blue)
p1b <- make_shape_plot(df_linlog, "Linear-log: Y = alpha + beta log(X)", green)
p1c <- make_shape_plot(df_loglin, "Log-linear: log(Y) = alpha + beta X", red)
p1d <- make_shape_plot(
  df_loglog,
  "Log-log: log(Y) = alpha + beta log(X)",
  purple,
  xl = "log(X)",
  yl = "log(Y)"
)

fig1 <- (p1a | p1b) / (p1c | p1d) +
  plot_annotation(
    title = "Four Logarithmic Transformation Models",
    subtitle = "Schematic curves based on Table 1 in Benoit (2011)",
    theme = theme(
```

```
    plot.title = element_text(face = "bold", size = 13, hjust = 0.5),
    plot.subtitle = element_text(size = 10, hjust = 0.5)
  )
)

fig1
```

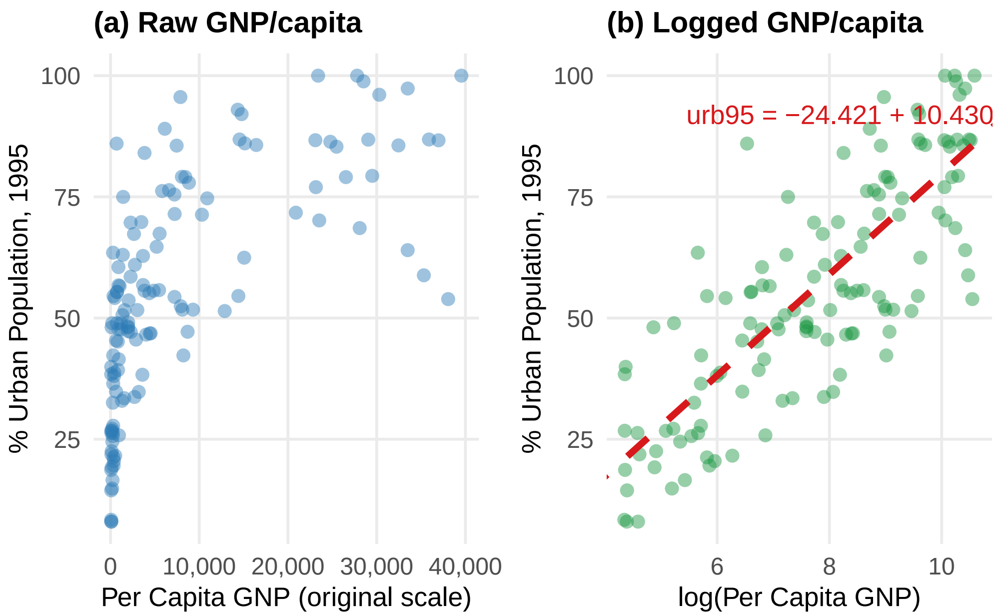



```
ggsave(
  file.path(out_dir, "fig1_four_models.png"),
  fig1,
  width = 10,
  height = 6.8,
  dpi = 150
)
```

Figure 2

```
set.seed(42)
n2 <- 132
log_gnp <- runif(n2, 4.34381, 10.6553)
urb95 <- pmax(
```

```r
  8,
  pmin(100, coef_linlog_intercept + coef_linlog_slope * log_gnp + rnorm(n2, 0, 15.646))
)
gnp_raw <- exp(log_gnp)

df2 <- data.frame(
  log_gnp = log_gnp,
  gnp_raw = gnp_raw,
  urb95 = urb95
)

label2 <- "urb95 = -24.421 + 10.430 * log(GNP/capita)"

p2a <- ggplot(df2, aes(gnp_raw, urb95)) +
  geom_point(alpha = 0.45, size = 1.8, color = blue) +
  scale_x_continuous(labels = label_number(big.mark = ",")) +
  labs(
    title = "(a) Raw GNP/capita",
    x = "Per Capita GNP (original scale)",
    y = "% Urban Population, 1995"
  ) +
  theme_paper

p2b <- ggplot(df2, aes(log_gnp, urb95)) +
  geom_point(alpha = 0.45, size = 1.8, color = green) +
  geom_abline(
    intercept = coef_linlog_intercept,
    slope = coef_linlog_slope,
    color = red,
    linetype = "dashed",
    linewidth = 1.2
  ) +
  annotate("text", x = 5.45, y = 92, label = label2, size = 3.4, color = red, hjust = 0) +
  labs(
    title = "(b) Logged GNP/capita",
    x = "log(Per Capita GNP)",
    y = "% Urban Population, 1995"
  ) +
  theme_paper

fig2 <- p2a | p2b
fig2
```

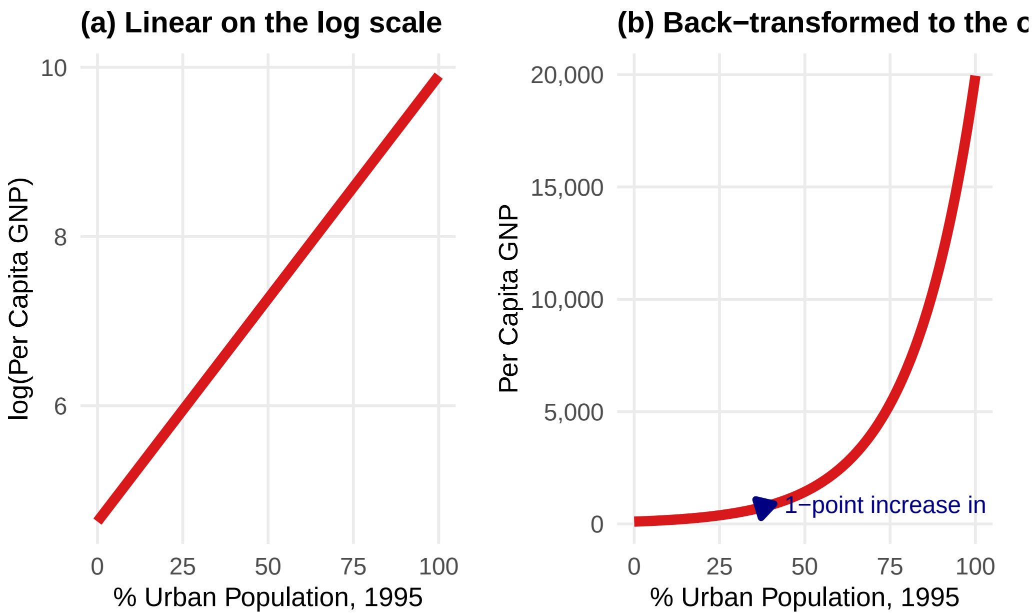



```r
ggsave(
  file.path(out_dir, "fig2_linear_log_example.png"),
  fig2,
  width = 10,
  height = 4.6,
  dpi = 150
)
```

Figure 3

```r
x3 <- seq(0, 100, length.out = 200)
logy3 <- coef_loglin_intercept + coef_loglin_slope * x3
y3 <- exp(logy3)

df3_log <- data.frame(x = x3, y = logy3)
df3_orig <- data.frame(x = x3, y = y3)

x_arr <- c(40, 41)
y_arr <- exp(coef_loglin_intercept + coef_loglin_slope * x_arr)
df_arr <- data.frame(
  x = x_arr[1],
  xend = x_arr[2],
```

```
  y = y_arr[1],
  yend = y_arr[2]
)

p3a <- ggplot(df3_log, aes(x, y)) +
  geom_line(color = red, linewidth = 1.8) +
  labs(
    title = "(a) Linear on the log scale",
    x = "% Urban Population, 1995",
    y = "log(Per Capita GNP)"
  ) +
  theme_paper

p3b <- ggplot(df3_orig, aes(x, y)) +
  geom_line(color = red, linewidth = 1.8) +
  geom_segment(
    data = df_arr,
    aes(x = x, xend = xend, y = y, yend = yend),
    arrow = arrow(length = unit(0.25, "cm"), type = "closed"),
    color = "navy",
    linewidth = 1.2
  ) +
  annotate(
    "text",
    x = 44,
    y = mean(y_arr),
    label = sprintf("1-point increase in urb95 multiplies GNP by e^beta = %.3f", exp(coef_log
    size = 3.3,
    color = "navy",
    hjust = 0
  ) +
  scale_y_continuous(labels = label_number(big.mark = ",")) +
  labs(
    title = "(b) Back-transformed to the original scale",
    x = "% Urban Population, 1995",
    y = "Per Capita GNP"
  ) +
  theme_paper

fig3 <- p3a | p3b
fig3
```

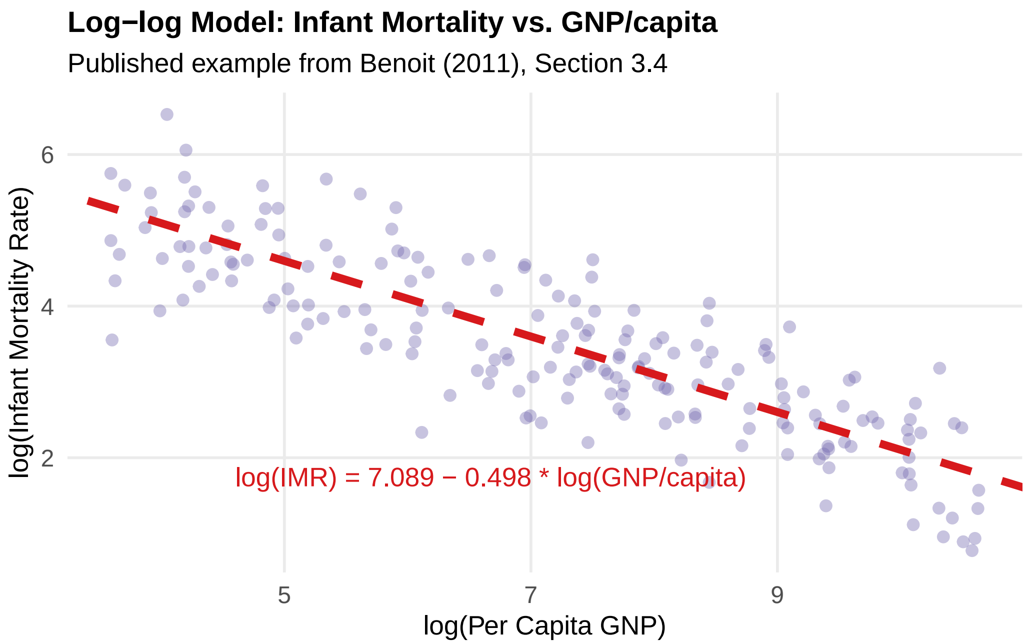



```
ggsave(
  file.path(out_dir, "fig3_loglinear_interp.png"),
  fig3,
  width = 10,
  height = 4.4,
  dpi = 150
)
```

Figure 4

```
set.seed(99)
n4 <- 194
log_gnp4 <- runif(n4, 3.58352, 10.6553)
log_imr <- coef_loglog_intercept + coef_loglog_slope * log_gnp4 + rnorm(n4, 0, 0.56915)
log_imr <- pmax(log(1), log_imr)

df4 <- data.frame(log_gnp = log_gnp4, log_imr = log_imr)

label4 <- "log(IMR) = 7.089 - 0.498 * log(GNP/capita)"

fig4 <- ggplot(df4, aes(log_gnp, log_imr)) +
  geom_point(alpha = 0.40, size = 1.6, color = purple) +
```

```
  geom_abline(
    intercept = coef_loglog_intercept,
    slope = coef_loglog_slope,
    color = red,
    linetype = "dashed",
    linewidth = 1.4
  ) +
  annotate("text", x = 4.6, y = 1.75, label = label4, size = 3.5, color = red, hjust = 0) +
  labs(
    title = "Log-log Model: Infant Mortality vs. GNP/capita",
    subtitle = "Published example from Benoit (2011), Section 3.4",
    x = "log(Per Capita GNP)",
    y = "log(Infant Mortality Rate)"
  ) +
  theme_paper

fig4
```




```
ggsave(
  file.path(out_dir, "fig4_loglog.png"),
  fig4,
  width = 7.4,
  height = 5.2,
```

```r
  dpi = 150
)
```

Saved Files

```r
cat("All 4 figures saved to:", normalizePath(out_dir), "\n")
```

```
All 4 figures saved to: /Users/sarah/Desktop/archeology/figures
```

```r
cat("Files:\n")
```

```
Files:
```

```r
cat("  fig1_four_models.png\n")
```

```
  fig1_four_models.png
```

```r
cat("  fig2_linear_log_example.png\n")
```

```
  fig2_linear_log_example.png
```

```r
cat("  fig3_loglinear_interp.png\n")
```

```
  fig3_loglinear_interp.png
```

```r
cat("  fig4_loglog.png\n")
```

```
  fig4_loglog.png
```